
ALGORITHM SPECIFICATION

NSE Intraday Trading Engine

A transparent, research-grade computational framework for generating actionable intraday intelligence on NSE Nifty 100 universe. Version 1.2 with corrected mathematics and verified statistical foundations.

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1. Executive Summary

This document specifies the corrected algorithm (Version 1.2) for the **NSE Intraday Trading Engine**, a research-grade computational framework that continuously monitors the NSE Nifty 100 universe during live trading sessions to produce ranked research coverage lists, detailed trade theses, and statistically grounded performance expectations.

Version 1.2 incorporates fourteen material corrections following adversarial mathematical audit of v1.1. The most significant fixes include: corrected factor weight normalization (now sums to exactly 100%), formally specified sub-factor composition formulas, mathematically attainable quality gates for the Hierarchical Edge Lookup, and elimination of contradictions between setup definitions and confluence requirements.

Engine Purpose Statement

The engine produces **research coverage** (not trade recommendations) consisting of Top 25 Long and Top 25 Short candidates, each with a thesis card showing trigger, target, stop-loss, net risk-reward after Indian transaction costs, actionability classification, and historically-calibrated expected value.

Key architectural properties:

- **No exclusion:** All 100 Nifty stocks are always scored and ranked; no stock is ever filtered out
- **No prediction claims:** The engine reports probabilistic expected values, not price forecasts
- **Honest cost accounting:** Every thesis includes itemized SEBI-schedule transaction costs
- **Actionability tiers:** Every thesis is classified as Tradeable, Constrained, or Research-Only
- **Self-calibrating edge statistics:** Historical performance data accumulates in a 6-tier hierarchical lookup

Scope Clarification

This document specifies algorithm logic only. Technology stack, data sources, and implementation architecture are covered in separate documents. The algorithm is intentionally stack-agnostic.

2. Engine Purpose and Design Philosophy

2.1 What the Engine Does

The engine operates continuously during the NSE trading session (9:15 AM to 3:30 PM IST) and performs the following functions every minute:

1. **Market identification (Layer 1):** Classifies the current market regime into one of three states: Trending-Up, Trending-Down, or Range-Bound, with volatility qualifier and time-of-day context.
2. **Stock identification (Layers 2-6):** Computes 7 factor scores for each of the 100 Nifty constituents, weights them by the current regime, applies liquidity and contextual modifiers, and ranks all stocks to produce Top 25 Long and Top 25 Short coverage lists.
3. **Thesis decoration (Layers 7-8):** Applies 6 mechanical confluence checks to each Top 25 entry, assembles trigger/target/stop levels, computes net risk-reward after full Indian cost accounting, applies time-decay, and classifies actionability.
4. **Outcome tracking (Layer 9):** Monitors every thesis from creation through terminal state, recording entry, exit, maximum favorable excursion, maximum adverse excursion, and realized return.
5. **Edge accumulation (Layer 10):** Aggregates historical outcomes into a 6-tier hierarchical lookup, providing statistically grounded expected value estimates that grow more granular as data accumulates.

2.2 Critical Design Principles

Principle 1: No Exclusion

No stock is ever removed from the scoring universe. Execution-path constraints (F&O ban, cash-only, poor liquidity) are attached as informational flags and reflected in the Actionability Tier, but the stock continues to be scored and ranked.

Principle 2: Honest Economics

Every thesis displays its economics *after* full Indian transaction costs (brokerage, STT, exchange charges, SEBI fees, stamp duty, GST, and depth-derived slippage). A setup with gross R:R of 2:1 but negative net R:R after costs is classified Research-Not-Tradeable.

Principle 3: Mechanical Only

Layer 7 confluence checks are purely deterministic and independently verifiable. No candlestick pattern recognition, no Elliott wave, no subjective interpretation. Every check produces a boolean pass/fail against unambiguous numerical thresholds.

Principle 4: Research-Only Product

The engine does not place orders, execute trades, or provide advice. It surfaces evidence; the trader exercises judgment. The Top 25 is research coverage, not a trade list. A typical user selects 3-5 candidates from the 50 based on personal criteria.

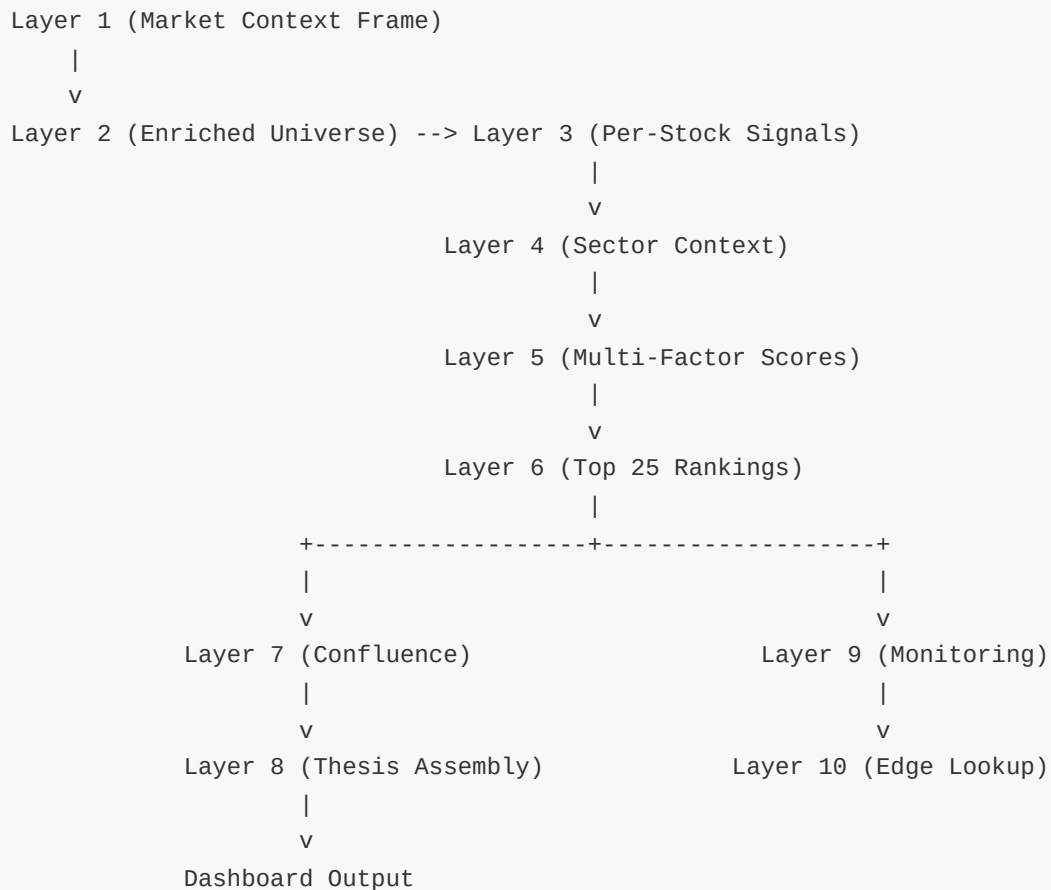
3. Layered Architecture Overview

The engine consists of 10 logical layers organized into three functional groups:

Table 1: Layer Groupings and Functions

Group	Layers	Function
Candidate Generation	L1 - L6	Produce ranked Top 25 Long and Top 25 Short
Thesis Decoration	L7 - L8	Attach confluence, economics, and actionability
Monitoring and Learning	L9 - L10	Track outcomes and accumulate edge statistics

The data flow between layers is strictly sequential:



Layers 1 through 6 execute on every 1-minute bar close. Layer 7 and the compact thesis (Layer 8 eager path) execute immediately after Layer 6. The full thesis (Layer 8 lazy path) generates on user request. Layer 9 monitors continuously on every tick. Layer 10 rebuilds nightly.

4. Layer 1: Market Context Layer

Purpose: Identify what kind of market today is. The Market Context Frame becomes the lens through which every stock is evaluated in Layer 5.

4.1 Inputs

Table 2: Layer 1 Inputs and Refresh Frequency

Input	Source	Frequency
Nifty 50 price, 5-min and 15-min bars	NSE Live	Real-time
Bank Nifty price and bars	NSE Live	Real-time
India VIX (current, percentile, 60-day trajectory)	NSE	1-min
Market breadth (% above VWAP, A/D ratio)	Computed	1-min
Time-of-day session bucket	System clock	Continuous
Event-day flag	Calendar	Daily pre-market
SGX/GIFT Nifty, Dow futures, Asian markets, USDINR, Brent	External	Pre-market only (6:00-9:15)

4.2 Regime Classification (3-State)

The 3-state regime classification replaces the 8-state system from v1.0. Three states is the empirically justified starting point; finer splits will be added only when 6+ months of outcome data proves they add predictive value.

Algorithm: Regime Classification

Step 1: Compute Nifty 50 log returns on 5-min bars: $r_t = \ln(P_t/P_{t-1})$

Step 2: Compute rolling 20-bar realized volatility: $\sigma_{20} = \sqrt{\frac{252 \times 78}{20} \sum_{i=t-19}^t (r_i - \bar{r})^2}$

Step 3: Compute volatility z-score against trailing 60-day 5-min baseline: $z_v = (\sigma_{20} - \mu_\sigma) / \sigma_\sigma$

Step 4: Compute 50-bar EMA slope on 5-min bars: $S = EMA_{50}(t) - EMA_{50}(t-5)$

Step 5: Classify regime:

Table 3: 3-State Regime Classification Rules

50-bar EMA Slope Condition	Regime Label
$S > +0.02\%$ per bar for at least 10 of last 15 bars	Trending-Up
$S < -0.02\%$ per bar for at least 10 of last 15 bars	Trending-Down
$ S \leq 0.02\%$ per bar (neither above condition met)	Range-Bound

Cold-Start Handling (First 90 Minutes)

At market open (9:15 AM), insufficient 5-min bars exist for a reliable 50-bar EMA. During 9:15-10:45 AM, use a *fast proxy regime*: compute the 9-bar EMA on 15-min bars and classify using the same slope thresholds. Transition to the primary 5-min 50-bar system at 10:45 AM (when 50 5-min bars have accumulated). The dashboard displays "Regime (Provisional)" during the proxy period.

Volatile qualifier: If $z_v > +1.5$, append "Volatile" qualifier. This does not create a new state; it modifies confidence in the regime classification and adjusts Layer 5 weights.

4.3 Volatility Environment

India VIX bands use **percentile-based** thresholds computed over trailing 90 trading days, not fixed absolute levels. Fixed levels decay as market structure evolves.

Table 4: VIX Percentile Bands (Dynamic)

Condition	Label	Implication
VIX < 20th percentile of trailing 90 days	Compressed	Mean-reversion favored; breakouts fragile
VIX between 20th and 80th percentile	Normal	Standard regime weights apply
VIX > 80th percentile of trailing 90 days	Elevated	Breakouts powerful; ranges unreliable

4.4 Time-of-Day Session Buckets

Table 5: Session Buckets and Characteristics

Window (IST)	Bucket	Characteristics
9:15-9:30	Opening Shock	Widest spreads, highest fakeout rate, avoid new entries
9:30-10:30	Trend Establishment	Strongest same-period persistence; primary decision window
10:30-11:30	Mid-Morning	Declining volatility; continuation setups favored
11:30-13:30	Lunch Lull	Lower variance; breakout failure rate elevated
13:30-14:30	Afternoon Recovery	Variance recovers; new setups possible
14:30-15:15	Square-Off	Last-hour variance ~1.5x session average; momentum amplifies
15:15-15:30	Closing Auction	Distorted by settlement flows; no new entries

4.5 Event Flags

The event flag is a label that feeds into Layer 8's Actionability Tier. On high-impact event days, certain setup types are automatically reclassified:

Table 6: Event Day Impact on Setup Types

Event	Impact
RBI Policy (11:00 AM)	Mean-reversion setups auto-classified Research-Only until 11:30 AM
Budget Day	All setups Constrained minimum; ORB expanded to 30-min
Election Results	Volatility qualifier forced to Elevated; range-bound weights suppressed
Weekly Expiry (Tue post-Sep 2025)	Options signals weighted 2x; futures signals unchanged
Earnings-Heavy (5+ stocks reporting)	Affected stocks earn -5 score penalty + Research-Only if during session

4.6 Market Breadth

Breadth is a composite of three metrics:

1. **VWAP participation:** Percentage of Nifty 100 trading above VWAP (primary signal)
2. **Advance/Decline ratio:** Nifty 100 stocks up vs. down vs. previous close
3. **Session high/low ratio:** Number of stocks making new session highs vs. lows

Breadth State Classification

Compute composite breadth $B = 0.5 \times VWAP_{pct} + 0.25 \times A_D_ratio_norm + 0.25 \times HL_ratio_norm$

Strong: $B > 0.60$ (trending day, broad participation)

Mixed: $0.40 \leq B \leq 0.60$

Weak: $B < 0.40$ (narrow, divergent, unreliable trend)

4.7 Pre-Market Global Cues

Global cues (SGX/GIFT Nifty, Dow futures, Asian markets, USDINR, Brent crude) are collected between 6:00-9:15 AM and synthesized into a single bias signal: Positive / Neutral / Negative.

Post-Open Decay

After 9:30 AM, global cue signals are ignored. By this time, overnight information has been absorbed into Nifty opening prices. Continuing to weight pre-market bias after the open introduces stale information.

4.8 Output: Market Context Frame

The Market Context Frame is refreshed every 5 minutes and contains the following fields:

```
{
  regime:                "Trending-Up" | "Trending-Down" | "Range-Bound"
  regime_confidence:     float [0.0, 1.0] (proxy=0.6, primary=0.85)
  volatility_qualifier:  "Volatile" | "Standard"
  vix_band:              "Compressed" | "Normal" | "Elevated"
  vix_trajectory:        "Rising" | "Stable" | "Falling"
  time_bucket:           one of 7 session buckets
  event_flag:            None | RBI | Budget | Election | Expiry | Earnings-Heavy
  breadth:               "Strong" | "Mixed" | "Weak"
  premarket_bias:        "Positive" | "Neutral" | "Negative" (valid 9:15-9:30 only)
  bank_nifty_divergence: float (Bank Nifty regime minus Nifty 50 regime, -2 to +2)
}
```

The `bank_nifty_divergence` field (new in v1.2) captures divergence between the two indices. When Bank Nifty and Nifty 50 disagree (e.g., Nifty trending up but Bank Nifty range-bound), this flag suppresses sector-specific bank setups and reduces Trend factor weight by 5 percentage points.

5. Layer 2: Universe Enrichment Layer

Purpose: Enrich all 100 Nifty constituents with informational flags and execution-path constraints. No stock is excluded.

5.1 Informational Flags

For each stock, the following flags are computed daily at 8:00 AM and updated intraday as needed:

Table 7: Layer 2 Informational Flags

Flag	Values	Source	Update
F&O Eligibility	F&O_ELIGIBLE / CASH_ONLY	NSE F&O list	Daily
F&O Ban	True / False	NSE Ban List	Daily
MWPL Proximity	None / Warning (>85%) / Critical (>92%)	NSE MWPL data	Daily
Circuit Proximity	None / Upper_2% / Lower_2%	Live price vs limits	1-min
Earnings Flag	None / Before_Market / During_Market / After_Market	NSE Corporate Actions	Daily
Index Change	None / Entered_Recently (<5 days) / Exiting_Soon	NSE Rebalance schedule	Event
Stale Data	True / False	Last tick timestamp	Tick

5.2 Liquidity Quality Score

The Liquidity Quality Score (LQS) is a composite metric computed from live order book data:

Liquidity Quality Score Computation

Step 1: Measure top-5 bid+ask depth at current price D (in Rs. lakhs)

Step 2: Measure bid-ask spread S as percentage of mid-price

Step 3: Get 10-day average daily turnover T (in Rs. crores)

Normalize each component to [0, 1] using percentile ranks against Nifty 100 universe:

$$D_{norm} = \text{percentile_rank}(D)$$

$$S_{norm} = 1 - \text{percentile_rank}(S) \text{ (lower spread = higher score)}$$

$$T_{norm} = \text{percentile_rank}(T)$$

$$\text{Composite: } LQS = 0.4 \times D_{norm} + 0.35 \times S_{norm} + 0.25 \times T_{norm}$$

Buckets: Excellent (LQS ≥ 0.80) / Good (0.55-0.79) / Marginal (0.30-0.54) / Poor (< 0.30)

5.3 Shortability Assessment

Shortability is critical for Indian markets because retail short-selling is heavily constrained. Four execution paths exist:

Table 8: Shortability Classification

Category	Execution Path	Constraints
Shortable via Futures/Options	F&O eligible, not in ban	Standard margin; carry overnight if desired
Shortable with Restrictions	F&O eligible, currently in ban	Existing positions only; no new short positions allowed
Shortable via MIS Only	Cash-only stock	Must square off by 3:20 PM; broker-specific MIS limits apply
Non-Shortable	Cash-only, no SLB availability	No viable short path for retail; appears only in Long thesis coverage

The shortability path feeds into Layer 8's Actionability Tier classification. A Non-Shortable stock can still appear in Top 25 Short (no exclusion), but its thesis is automatically Research-Only.

6. Layer 3: Per-Stock Signal Layer

Purpose: Compute the full technical state of every stock as of the most recent bar close. All signals are refreshed every minute.

6.1 Trend Indicators

Table 9: Trend Indicator Specifications

Indicator	Parameters	Timeframe	Output
EMA Stack	EMA(9), EMA(20), EMA(50)	5-min and 15-min	Binary: aligned (bullish/bearish) or mixed
Supertrend	ATR(10) x 3.0 multiplier	5-min only	Direction + distance to flip
ADX	ADX(14)	5-min and 15-min	Value 0-100; >25 = trending

Supertrend Parameters (Fixed)

Supertrend uses ATR period = 10 and multiplier = 3.0 on 5-min bars. These are fixed across all stocks. No per-stock optimization. The 3.0 multiplier is chosen to reduce whipsaw frequency; it produces fewer signals but higher reliability than the common 2.0 setting.

6.2 Momentum Indicators

Table 10: Momentum Indicator Specifications

Indicator	Parameters	Timeframe	Output
RSI	RSI(14)	5-min and 15-min	Value 0-100; <30 oversold, >70 overbought
MACD Histogram	EMA(12,26,9)	15-min only	Value + direction (rising/falling)
ROC vs Nifty	20-bar rate of change	5-min	Stock return minus Nifty return

MACD Divergence — Formal Definition

MACD divergence detection is *not* pattern-based. It is defined mechanically as follows:

Bullish divergence (for Long setups): Price registers a 5-bar lower low (current close < previous 4-bar low) AND MACD histogram registers a 5-bar higher low (current histogram > minimum of previous 4 bars).

Bearish divergence (for Short setups): Price registers a 5-bar higher high (current close > previous 4-bar high) AND MACD histogram registers a 5-bar lower high (current histogram < maximum of previous 4 bars).

Both conditions must be confirmed by bar close. No human interpretation required.

6.3 Volatility Indicators

Table 11: Volatility Indicator Specifications

Indicator	Parameters	Timeframe	Output
ATR	ATR(14)	5-min and 15-min	Absolute ATR value
ATR Percentile	ATR vs 20-day distribution	5-min	Percentile rank of current ATR
Bollinger Bands	SMA(20) +/- 2*sigma	5-min	Band width + position within band
Realized vs Implied	10-day realized vol / ATM IV	Daily	RV/IV ratio; <1 = options overpricing vol

6.4 Volume Context

Volume is adjusted for intraday seasonality before computing signals:

Seasonally-Adjusted Volume Computation

Step 1: For each stock, compute average volume for each 5-minute interval over trailing 10 days. This produces a seasonal profile $V_{seasonal}(t)$ where t is the time-of-day bucket.

Step 2: Current raw volume V_{raw} is divided by the seasonal factor:

$$V_{adj} = V_{raw} / V_{seasonal}(t)$$

Step 3: Volume z-score uses adjusted volume:

$$z_v = (V_{adj} - \mu_{V_{adj}}) / \sigma_{V_{adj}} \text{ over trailing 10 days}$$

Step 4: Volume confirmation flag (used in Layer 7):

$$V_{confirm} = V_{adj} \geq 1.5 \times \text{median adjusted volume}$$

Additional volume signals:

- **VWAP position:** Current price relative to VWAP (above, at, below) and distance to +/-1sigma and +/-2sigma bands
- **Volume profile:** Distribution of volume across price levels for the current session

6.5 Reference Levels

Reference levels are computed once at market open and remain fixed for the session:

Table 12: Reference Levels Computed at 9:15 AM

Level	Computation
Previous Day High (PDH)	Yesterday's session high
Previous Day Low (PDL)	Yesterday's session low
Previous Day Close (PDC)	Yesterday's closing price
Central Pivot (CP)	$(PDH + PDL + PDC)/3$
Bottom Central (BC)	$(PDH + PDL)/2$
Top Central (TC)	$CP - BC + CP = 2 \times CP - BC$
Floor R1	$2 \times CP - PDL$
Floor R2	$CP + (PDH - PDL)$
Floor R3	$PDH + 2 \times (CP - PDL)$
Floor S1	$2 \times CP - PDH$
Floor S2	$CP - (PDH - PDL)$
Floor S3	$PDL - 2 \times (PDH - CP)$
ORB 15-min High	Highest price 9:15-9:30 AM
ORB 15-min Low	Lowest price 9:15-9:30 AM
ORB 2-hour High	Highest price 9:15-11:15 AM
ORB 2-hour Low	Lowest price 9:15-11:15 AM
First Hour High	Highest price 9:15-10:15 AM
First Hour Low	Lowest price 9:15-10:15 AM

6.6 Open Interest Classification

For F&O eligible stocks only. OI change is computed over a 15-minute smoothing window to reduce noise:

Table 13: Open Interest Classification

Price Change (15-min)	Open Interest Change (15-min)	Classification
Rising ($> +0.5\%$)	Increasing ($> +2\%$)	Long Buildup — bullish conviction
Falling ($< -0.5\%$)	Increasing ($> +2\%$)	Short Buildup — bearish conviction
Falling ($< -0.5\%$)	Decreasing ($< -2\%$)	Long Unwinding — longs exiting
Rising ($> +0.5\%$)	Decreasing ($< -2\%$)	Short Covering — shorts exiting
Neither	Either	Neutral — no clear OI signal

6.7 Options-Derived Signals

For F&O eligible stocks only:

Table 14: Options-Derived Signals

Signal	Computation
IV Percentile	Current ATM IV vs trailing 60-day distribution (0-100%)
IV Percentile (Long)	Current ATM IV vs trailing 1-year distribution (0-100%)
Expected Range	$\text{ATM price} \pm (\text{IV} / \sqrt{252}) \times \text{ATM price}$
PCR Z-Score (Short)	$(\text{PCR} - 20\text{d mean}) / 20\text{d stdev}$; extreme values signal contrarian opportunities
PCR Z-Score (Long)	$(\text{PCR} - 1\text{yr mean}) / 1\text{yr stdev}$; structural context
RV/IV Ratio	10-day realized vol / ATM implied vol; < 1 = options expensive

7. Layer 4: Sector Context Layer

Purpose: Position each stock against its sector and the sector against the market.

There are 11 NSE sectoral indices tracked: Auto, Bank, FMCG, IT, Media, Metal, Pharma, PSU Bank, Realty, Energy, and Telecom. The Financial Services sector is decomposed into Bank and PSU Bank for granularity.

7.1 Sector Relative Strength

Sector RS Computation

Step 1: Compute 5-day and 20-day returns for each sector index and Nifty 50.

Step 2: RS-Ratio (normalized relative strength):

$$RSR_i = \frac{R_{sector,i}^{5d} / R_{Nifty}^{5d}}{60\text{-day rolling std dev of same ratio}}$$

Step 3: RS-Momentum (rate of change of RS-Ratio):

$$RSM_i = RSR_i(t) - RSR_i(t - 5) \text{ (5-day change in RS-Ratio)}$$

Step 4: Rank all 11 sectors by RSR. Track rank change over last 30 minutes.

Step 5: Sector rotation classification:

- Gaining: rank up by 2+ positions in 30 min
- Steady: rank unchanged +/-1
- Losing: rank down by 2+ positions in 30 min

Every stock inherits its sector's current rank, RS-Ratio, RS-Momentum, and rotation state. This sector context feeds into Layer 5's Sector Factor.

8. Layer 5: Multi-Factor Scoring Layer

Purpose: Combine all signals into a Long Score and Short Score per stock, weighted by the current market regime. This is the mathematical core of the engine.

8.1 The Seven Factor Sub-Scores

Each stock receives seven factor sub-scores, each on a 0-100 scale. The factors are:

Table 15: Seven Factor Sub-Scores

#	Factor	Components
F1	Trend Factor	EMA stack alignment, Supertrend direction, ADX strength
F2	Momentum Factor	RSI position, MACD histogram divergence (mechanical), ROC vs Nifty
F3	Volume Factor	Seasonally-adjusted volume vs 10-day avg, VWAP position, volume z-score
F4	Volatility-Position Factor	ATR-normalized distance from key levels, BB position, ATR percentile
F5	Sector Factor	Sector RS-Ratio + RS-Momentum
F6	Open Interest Factor	OI buildup classification (F&O only; CASH_ONLY gets neutral 50)
F7	Position-in-Range Factor	52-week range position, distance to CPR/PDH/PDL

8.2 Factor Composition Formulas

Each factor is computed as a weighted average of its components, normalized to [0, 100]. Component scores are computed as follows:

F1: Trend Factor Composition

Components (all boolean, scored 0 or 100):

- C1: $\text{EMA}(9) > \text{EMA}(20) > \text{EMA}(50)$ on 5-min (bullish) [100 if true, 0 if false]
- C2: $\text{EMA}(9) > \text{EMA}(20) > \text{EMA}(50)$ on 15-min (bullish) [100 if true, 0 if false]
- C3: Supertrend direction = bullish [100 if true, 0 if false]
- C4: $\text{ADX}(14)$ on 5-min > 25 [$\min(100, \text{ADX} \times 4)$]
- C5: $\text{ADX}(14)$ on 15-min > 25 [$\min(100, \text{ADX} \times 4)$]

$$F1 = 0.25 \cdot C1 + 0.25 \cdot C2 + 0.20 \cdot C3 + 0.15 \cdot C4 + 0.15 \cdot C5$$

For Short Score: invert C1, C2, C3 (bearish alignment = 100)

F2: Momentum Factor Composition

- C1: RSI position [100 if RSI in 40-60 neutral zone for range-bound; 100 if $\text{RSI} > 55$ for trending-up; 100 if $\text{RSI} < 45$ for trending-down]
- C2: MACD histogram divergence (mechanical definition from 6.2) [100 if divergence confirmed, 0 otherwise]
- C3: ROC vs Nifty z-score [100 if $\text{ROC } z > +0.5$ for long; 100 if $\text{ROC } z < -0.5$ for short]

$$F2 = 0.30 \cdot C1 + 0.35 \cdot C2 + 0.35 \cdot C3$$

F3: Volume Factor Composition

- C1: Seasonally-adjusted volume ratio [$\min(100, \text{ratio} \times 50)$]
- C2: VWAP position [100 if price > VWAP + 0.5*sigma for long; 100 if price < VWAP - 0.5*sigma for short]
- C3: Volume z-score [$\min(100, \max(0, (z+2)*25))$]

$$F3 = 0.40*C1 + 0.30*C2 + 0.30*C3$$

F4: Volatility-Position Factor Composition

- C1: ATR-normalized distance to nearest support (for long) or resistance (for short) [100 if distance > 1.5x ATR]
- C2: Bollinger Band position [100 if price between 0-sigma and -1.5-sigma for long; 100 if between 0-sigma and +1.5-sigma for short]
- C3: ATR percentile [100 if ATR percentile > 60 (sufficient volatility for moves)]

$$F4 = 0.40*C1 + 0.35*C2 + 0.25*C3$$

F5: Sector Factor Composition

- C1: Sector RS-Ratio normalized [$\max(0, \min(100, (\text{RSR} + 2) * 25))$]
- C2: Sector RS-Momentum [$\max(0, \min(100, (\text{RSM} + 1) * 50))$]

$$F5 = 0.60*C1 + 0.40*C2$$

F6: Open Interest Factor Composition

For F&O eligible stocks:

- Long Buildup = 100
- Short Covering = 70
- Neutral = 50
- Long Unwinding = 30
- Short Buildup = 0 (inverted for Short Score: Short Buildup = 100, Long Buildup = 0)

For CASH_ONLY stocks: F6 = 50 (neutral, neither helps nor hurts)

F7: Position-in-Range Factor Composition

- C1: 52-week range position [100 if in bottom 20% for long; 100 if in top 20% for short]
- C2: Distance to CPR [100 if price near CPR with room to move toward PDH/PDL]
- C3: ORB position [100 if price near ORB low for long breakout; 100 if near ORB high for short breakout]

$$F7 = 0.35*C1 + 0.35*C2 + 0.30*C3$$

8.3 Regime-Conditional Weighting

The seven factors are combined using regime-conditional weights. **Weights sum to exactly 100%** per regime (corrected from v1.1):

Table 16: Regime-Conditional Factor Weights (Sum = 100%)

Regime	F1 Trend	F2 Mom	F3 Vol	F4 Vol-Pos	F5 Sector	F6 OI	F7 Pos-Rng
Trending-Up	25%	20%	12%	5%	18%	12%	8%
Trending-Down	25%	20%	12%	5%	18%	12%	8%
Range-Bound	8%	5%	18%	30%	15%	12%	12%

Weight Design Rationale

In trending regimes, trend and momentum dominate (45% combined). In range-bound regimes, volatility-position and volume dominate (48% combined), while trend and momentum are suppressed to 13% combined. This reflects the empirical reality that trend-following works in trending markets and mean-reversion works in ranging markets.

8.4 Liquidity Quality Multiplier

After computing the weighted composite, a liquidity multiplier is applied:

Table 17: Liquidity Quality Multiplier

Liquidity Quality	Multiplier	Rationale
Excellent	1.00	Negligible slippage expected
Good	0.97	Minor slippage drag (~3 bps)
Marginal	0.92	Material slippage (~8 bps)
Poor	0.85	Severe slippage (~15 bps)

This is a *soft penalty* — poor-liquidity stocks can still rank Top 25 if their signal is strong, but their score reflects expected execution drag.

8.5 Contextual Modifiers

After the liquidity multiplier, contextual modifiers are applied additively:

Table 18: Contextual Score Modifiers

Condition	Modifier	Rationale
F&O Ban flag	-4 points	Informational; doesn't exclude
Earnings during session	-6 points + variance warning	Elevated event risk
Stale data (>30 sec)	Score frozen at last known	Prevents trading on stale signals
Strong sector tailwind	+3 points	Sector momentum amplification
Weak sector headwind	-3 points	Sector drag
Index constituent change (<5 days)	-2 points	Artificial volume/price pressure

8.6 Long Score and Short Score Computation

The complete scoring algorithm:

Algorithm: Score Computation

Step 1: Compute 7 factor scores F_1 - F_7 for each stock (0-100 each).

Step 2: Apply regime weights w_i from Table 16:

$$S_{raw} = \sum_{i=1}^7 w_i \times F_i$$

Step 3: Apply liquidity multiplier m_L :

$$S_{liq} = S_{raw} \times m_L$$

Step 4: Apply contextual modifiers (additive):

$$S_{final} = S_{liq} + \sum modifiers$$

Step 5: Clamp to [0, 100]:

$$S_{final} = \max(0, \min(100, S_{final}))$$

Step 6: Compute Long Score = S_{final} with bullish component interpretations.

Step 7: Compute Short Score = S_{final} with bearish component interpretations, then apply asymmetry penalty:

$$ShortScore_{effective} = ShortScore_{raw} \times 0.92$$

The 0.92 asymmetry factor reflects empirical evidence that false-short rate exceeds false-long rate in NSE intraday data. Short setups require higher raw confidence to achieve the same effective score.

9. Layer 6: Cross-Sectional Ranking Layer

Purpose: Produce Top 25 Long and Top 25 Short from the scored universe. No filtering, no gates, no exclusion.

9.1 Hysteresis Mechanism

Hysteresis prevents rank flickering that would degrade dashboard UX. The threshold is adaptive:

Adaptive Hysteresis Algorithm

Threshold computation (updated every 5 minutes):

$\sigma_{gap} = \text{std dev of score gaps between ranks 20-30}$

$\theta = \max(2.0, 0.25 \times \sigma_{gap})$

Entry rule: A stock at rank 26+ enters Top 25 only if its score exceeds current rank 25's score by at least θ .

Exit rule: A stock at rank 25 drops out only if rank 26's score exceeds it by at least θ .

Target: ~1 rank change per minute on average. If observed rate exceeds 2/min, increase θ by 10%. If below 0.5/min, decrease θ by 10%.

9.2 Rank Movement Tracking

Each entry in Top 25 carries movement indicators:

- **NEW:** Entered Top 25 within last 5 minutes
- **UP:** Rank improved by 2+ positions in last 5 minutes
- **DOWN:** Rank declined by 2+ positions in last 5 minutes
- **STABLE:** Rank unchanged (+/-1 position)

9.3 Concentration Metrics

Informational displays (no filtering applied):

Table 19: Concentration Metrics

Metric	Computation	Interpretation
Sector Concentration	Max count of any single sector in Top 25	>8 suggests theme-driven day; diversify carefully
Score Spread	Score(rank 1) - Score(rank 25)	>20 = strong conviction; <10 = uncertain market
Correlation Cluster	Count of pairs with cosine similarity > 0.70	>15 pairs suggests crowded factor exposure

Correlation cluster detection uses cosine similarity of the 7-dimensional factor score vector between pairs of stocks in Top 25. Two stocks with factor profiles pointing in the same direction are likely to move together.

10. Layer 7: Mechanical Confluence Layer

Purpose: Compute a Confluence Score (0-6) using six deterministic, mechanical checks. No subjective pattern recognition.

10.1 The Six Deterministic Checks

Table 20: Six Mechanical Confluence Checks

#	Check	Definition	Pass Condition
1	Strong Close	Trigger bar close position within its range	Close in upper 33% of H-L range (long) / lower 33% (short)
2	Volume Confirmation	Trigger bar volume vs recent average	Seasonally-adjusted volume $\geq 1.5 \times$ 20-bar median
3	Non-Exhaustion	Trigger bar range vs recent median	Bar range $\leq 1.5 \times$ 20-bar median range (avoids climax)
4	Higher-TF Alignment	15-min EMA stack direction	EMA(9) > EMA(20) > EMA(50) on 15-min (long); inverse (short)
5	Adequate Risk Distance	Room before invalidation	$ \text{Price} - \text{Invalidation} \geq 0.5 \times \text{ATR}(14, 5\text{-min})$
6	Adequate Reward Distance	Reward vs risk ratio	$ \text{T1} - \text{Price} \geq 1.2 \times \text{Price} - \text{Invalidation} $ (gross R:R ≥ 1.2)

Check 6 Threshold: 1.2 (not 1.5)

v1.1 used 1.5, which filtered many valid setups that were net-positive after low-cost execution (high-liquidity stocks). The 1.2 threshold allows more setups through to Layer 8, where the full cost model determines true viability. Layer 8's Net R:R ≥ 1.0 requirement provides the final filter.

10.2 Time-of-Day Adjustments

Check 2 (Volume Confirmation) is adjusted for intraday seasonality using the same seasonal profile as Layer 3's volume computation. During Opening Shock (9:15-9:30), the volume threshold is raised to 2.0x (from 1.5x) to account for the natural opening volume spike.

11. Layer 8: Thesis Assembly Layer

Purpose: For each Top 25 entry, build a thesis card with honest economics and clear actionability classification.

11.1 Setup Type Definitions (All Six Types)

v1.1 defined only 4 setup types but Layer 10 tracked 6. v1.2 formally defines all 6 setup types with complete parameter tables:

Table 21: Setup Type 1: Opening Range Breakout (ORB)

Parameter	Long	Short
Trigger	ORB High + 1 tick	ORB Low - 1 tick
Invalidation	max(ORB Low, VWAP - 0.5%)	min(ORB High, VWAP + 0.5%)
T1	Trigger + 1.5 x ORB Range	Trigger - 1.5 x ORB Range
T2	Previous Day High	Previous Day Low
Valid Window	Until 11:00 AM (15-min ORB) or 13:00 (2-hour ORB)	Same
Preferred Regime	Trending-Up, Normal VIX	Trending-Down, Normal VIX

Table 22: Setup Type 2: VWAP Reclaim

Parameter	Long	Short
Trigger	VWAP cross above with volume confirmation	VWAP cross below with volume confirmation
Invalidation	VWAP - 0.8 x ATR(14, 5m)	VWAP + 0.8 x ATR(14, 5m)
T1	VWAP + 1.5 x ATR	VWAP - 1.5 x ATR
T2	VWAP + 2.5 x ATR	VWAP - 2.5 x ATR
Valid Window	After 9:45 AM only	After 9:45 AM only
Preferred Regime	Trending-Up, Normal/Elevated VIX	Trending-Down, Normal/Elevated VIX

Table 23: Setup Type 3: Supertrend Pullback

Parameter	Long	Short
Trigger	Pullback touches Supertrend line from above	Pullback touches Supertrend line from below
Invalidation	Supertrend line - 0.5 x ATR	Supertrend line + 0.5 x ATR
T1	Supertrend line + 1.5 x ATR	Supertrend line - 1.5 x ATR
T2	Supertrend line + 2.5 x ATR	Supertrend line - 2.5 x ATR
Valid Window	Any time Supertrend is bullish	Any time Supertrend is bearish
Preferred Regime	Trending-Up	Trending-Down

Table 24: Setup Type 4: Mean Reversion at Band

Parameter	Long	Short
Trigger	Price touches lower 2-sigma Bollinger Band	Price touches upper 2-sigma Bollinger Band
Invalidation	Band breach below 2.5-sigma	Band breach above 2.5-sigma
T1	VWAP (must exceed trigger by $\geq 0.6 \times$ invalidation distance)	VWAP (must exceed trigger by $\geq 0.6 \times$ invalidation distance)
T2	Opposite 1-sigma band	Opposite 1-sigma band
Valid Window	Any time; best in Range-Bound regime	Any time; best in Range-Bound regime
Preferred Regime	Range-Bound, Compressed VIX	Range-Bound, Compressed VIX

Mean Reversion T1 Correction

v1.1's Mean Reversion setup had T1 at VWAP, which often produced gross R:R < 1.0 , violating Check 6. In v1.2, T1 for Mean Reversion is adjusted to ensure $|T1 - \text{Trigger}| \geq 0.6 \times |\text{Trigger} - \text{Invalidation}|$, guaranteeing gross R:R ≥ 1.2 when combined with Check 6's 1.2 threshold. If VWAP is closer than this minimum, T1 is set to the minimum viable distance.

Table 25: Setup Type 5: First Hour Breakout

Parameter	Long	Short
Trigger	Break above First Hour High after 10:15 AM	Break below First Hour Low after 10:15 AM
Invalidation	First Hour Low - 0.3 x FH Range	First Hour High + 0.3 x FH Range
T1	Trigger + 1.0 x FH Range	Trigger - 1.0 x FH Range
T2	PDH	PDL
Valid Window	10:15 AM - 12:00 PM	10:15 AM - 12:00 PM
Preferred Regime	Trending-Up	Trending-Down

Table 26: Setup Type 6: CPR Breakout

Parameter	Long	Short
Trigger	Break above TC (Top Central) with volume	Break below BC (Bottom Central) with volume
Invalidation	BC - 0.2 x CPR Width	TC + 0.2 x CPR Width
T1	R1 Floor Pivot	S1 Floor Pivot
T2	R2 Floor Pivot	S2 Floor Pivot
Valid Window	After 9:45 AM	After 9:45 AM
Preferred Regime	Trending-Up	Trending-Down

11.2 Indian Cost Model

The cost model is itemized per-leg (entry and exit) using current SEBI/NSE schedules as of May 2026.

Equity Intraday (Cash, MIS)

Table 27: Equity Intraday Cost Schedule

Charge	Rate	Side
Brokerage	0.03% or Rs. 20/order, whichever is lower	Both
STT	0.025%	Sell only
Exchange Transaction	0.00297% (NSE)	Both
SEBI Charges	Rs. 10/crore = 0.0001%	Both
Stamp Duty	0.003% (Maharashtra)	Buy only
GST	18% on (brokerage + exchange + SEBI)	Both

Equity Futures Intraday

Table 28: Futures Intraday Cost Schedule

Charge	Rate	Side
Brokerage	Rs. 20/leg flat (Zerodha-style)	Both
STT	0.0125% on sell side	Sell only
Exchange Transaction	0.00173% (NSE futures)	Both
SEBI Charges	Rs. 10/crore = 0.0001%	Both
Stamp Duty	0.002%	Buy only
GST	18% on (brokerage + exchange + SEBI)	Both

Slippage Model (Depth-Derived)

Table 29: Slippage by Liquidity Quality

Liquidity	Normal Slippage/Leg	Stop-Loss Slippage/Leg
Excellent	5 bps (0.05%)	+8 bps (13 bps total)
Good	10 bps (0.10%)	+15 bps (25 bps total)
Marginal	20 bps (0.20%)	+25 bps (45 bps total)
Poor	35 bps (0.35%)	+40 bps (75 bps total)

Cost Drag Display

The thesis card explicitly displays: "Costs consume X% of your Y% target." For example, at Rs. 1,00,000 ticket with Good liquidity: round-trip costs ~Rs. 110 (0.11%), so a 0.5% target yields 0.39% net. The dashboard shows: "Costs: 0.11% drag on 0.50% target = 22% of gross captured."

11.3 Net Risk-Reward Computation

Algorithm: Net R:R Computation

Step 1: Gross R:R = $|T1 - \text{Trigger}| / |\text{Trigger} - \text{Invalidation}|$

Step 2: Round-trip cost % = Sum of all charges (Table 27 or 28) x 2 (entry + exit) + slippage x 2

Step 3: Net reward distance = $|T1 - \text{Trigger}| - (\text{round-trip cost \%} \times \text{Trigger price})$

Step 4: Net risk distance = $|\text{Trigger} - \text{Invalidation}| + (\text{stop-slippage} \times \text{Trigger price})$

Step 5: Net R:R = Net reward distance / Net risk distance

Quality Grade:

- Net R:R ≥ 1.5 : ATTRACTIVE (Green)
- $1.0 \leq \text{Net R:R} < 1.5$: MARGINAL (Amber)
- Net R:R < 1.0 : UNATTRACTIVE - Costs exceed edge (Red)

11.4 Time-Decay Function

The time-decay multiplier uses a continuous exponential decay function rather than v1.1's step function:

$$M(t) = \exp(-\lambda \cdot \max(0, t - t_{\text{window}})^2) \quad (1)$$

Where:

- t = current time (minutes since 9:15 AM)
- t_{window} = optimal window start time for the setup type (in minutes)
- λ = decay constant = 0.0003 for ORB setups, 0.00015 for Supertrend/VWAP setups

The practical effect at key times:

Table 30: Time-Decay Multiplier at Key Times

Time (IST)	ORB Multiplier	Supertrend/VWAP Multiplier
9:30-10:30	1.00	1.00
10:30-11:30	0.85	0.90
11:30-12:30	0.65	0.78
12:30-13:30	0.42	0.62
13:30-14:30	0.22	0.42
After 14:30	0.08	0.22

Displayed in thesis: "If triggered now: EV +0.34%. Time-adjusted EV: +0.20%."

11.5 Actionability Tier Classification

Every thesis is classified into one of three tiers:

Table 31: Actionability Tier Classification

Tier	Criteria	Visual Treatment
Tradeable	Clean execution path AND Net R:R ≥ 1.0 AND liquidity $\geq$ Good AND no blocking flags (F&O ban for futures, earnings during session)	Full color, full prominence, green Net R:R badge
Constrained	Execution path exists but with material friction: F&O ban with workaround, Marginal liquidity, Cash-only with MIS required, or Net R:R 0.8-1.0	Standard display with yellow warning labels, explicit reason shown
Research-Only	No viable retail execution: Cash-only for short with no SLB, Net R:R < 0.8 , Poor liquidity, or setup invalidated by event flag	Greyed out, deprioritized, labeled "Research-Only" with specific reason

11.6 Compact and Full Thesis Structure

Compact Thesis (Eager, Always Visible)

```

+-----+
| TATAMOTORS | LONG | Tradeable |
|            |      |          |
| Net R:R: 1.4:1 [ATTRACTIVE] |
| Setup: ORB Long | Confluence: 5/6 | Score: 87 |
|            |      |          |
| Trigger: Rs. 715.00 |
| SL: Rs. 708.50 (-0.91%) |
| T1: Rs. 722.00 (+0.98%) | T2: Rs. 728.00 |
|            |      |          |
| Time-adjusted EV: +0.20% |
| Cost drag: 0.11% (22% of gross target) |
| [+] Expand for full thesis |
+-----+

```

Full Thesis (Lazy, On Expand)

When expanded, the thesis card includes:

1. **The Case:** 6 mechanical Confluence checks with current values and pass/fail
2. **Cost breakdown:** Itemized brokerage, STT, exchange, SEBI, stamp, GST, slippage — total and % of target
3. **Live Invalidation Monitors:** 5-7 conditions tracked every minute (e.g., "Price below VWAP" = invalidated)
4. **Historical Edge:** From Layer 10's hierarchical lookup, with tier label and sample size
5. **Multi-Timeframe Trend:** Daily / 15-min / 5-min trend status
6. **Reference Levels:** CPR, Floor Pivots, ORB (15-min and 2-hour), First Hour H/L
7. **Context Flags:** F&O ban, MWPL proximity, circuit proximity, earnings, IV percentile
8. **Why This Setup:** Brief explanation of the setup logic (e.g., "Opening range breakout with volume confirmation in a trending market")

12. Layer 9: Outcome Monitoring Layer

Purpose: Track every thesis from creation through terminal state. Force-expire at end of intraday session.

12.1 State Machine

```

CREATED ----> PENDING ----> TRIGGERED ----> ACTIVE
      |               |               |
      |               |               +--> T1_HIT
      |               |               +--> T2_HIT
      |               |               +--> STOPPED_OUT
      |               |               +--> INVALIDATED
      |               |               +--> EXPIRED (conditional)
      |               +--> (skipped if no trigger by expiry time)
      |
      +--> FORCE_EXPIRED at 15:15 IST

```

12.2 Per-Outcome Metrics

For every thesis that reaches terminal state:

Table 32: Per-Outcome Metrics

Metric	Definition
Entry Timestamp	When trigger price was first crossed
Exit Timestamp	When terminal state was reached
MFE (Max Favorable Excursion)	Best price reached after entry, as % of entry
MAE (Max Adverse Excursion)	Worst price reached after entry, as % of entry
MFE Timestamp	Time of maximum favorable excursion
Gross Return	$(\text{Exit} - \text{Entry}) / \text{Entry}$ as percentage
Net Return	Gross Return - round-trip cost %
R-Multiple	$\text{Net Return} / \text{Entry} - \text{Invalidation} $
Time-to-Trigger	Minutes from thesis creation to trigger
Time-to-Exit	Minutes from trigger to terminal state

12.3 Session Boundaries

Expiry rules are setup-type dependent (corrected from v1.1's blanket 14:00 rule):

Table 33: Session Expiry Rules by Setup Type

Setup Type	Pending Expiry	Force Expiry	Rationale
ORB (15-min)	11:00 AM	15:15 IST	ORB invalidates if not triggered within ~90 min
ORB (2-hour)	13:00 PM	15:15 IST	2-hour ORB has wider window
VWAP Reclaim	14:00 PM	15:15 IST	VWAP setups valid through afternoon
Supertrend Pullback	14:30 PM	15:15 IST	Supertrend valid until square-off
Mean Reversion	13:30 PM	15:15 IST	Mean reversion decays fastest
First Hour Breakout	12:00 PM	15:15 IST	FH setups must trigger by noon
CPR Breakout	14:00 PM	15:15 IST	CPR setups valid through afternoon

Conditional Extension

If volatility is recovering (VIX rising from lunch lows and 5-min realized vol > 80th percentile of session), pending expiry is extended by 30 minutes for VWAP and Supertrend setups. This captures post-lunch momentum recovery that v1.1's blanket 14:00 rule missed.

13. Layer 10: Hierarchical Edge Lookup Layer

Purpose: Accumulate honest performance evidence and present it usefully from Day 1. This is the most important architectural component for long-term engine value.

13.1 The Six-Tier Fallback Structure

For each thesis, the system walks from most-specific to most-general, returning the most specific tier that passes its quality gate:

Table 34: Hierarchical Edge Lookup Tiers

Tier	Aggregation	Approx. Cells	Quality Gate	Timeline
Tier 1	Setup x Regime x Sector x Time-bucket	1,386 (6x3x11x7)	n >= 30, CI half-width <= 15%	Year 2+
Tier 2	Setup x Regime x Time-bucket	126 (6x3x7)	n >= 40, CI half-width <= 14%	Month 6+
Tier 3	Setup x Regime	18 (6x3)	n >= 50, CI half-width <= 14%	Month 3+
Tier 4	Setup x Time-bucket	42 (6x7)	n >= 50, CI half-width <= 14%	Month 2+
Tier 5	Setup baseline	6	n >= 80, CI half-width <= 12%	Month 1+
Tier 6	Global baseline	1	n >= 50, CI half-width <= 14%	Week 1+

Tier 1 Cell Count Correction

v1.1 stated ~3,700 cells for Tier 1. With 6 setups x 3 regimes x 11 sectors x 7 time-buckets = 1,386 cells (2,772 if Long/Short counted separately). v1.2 corrects this and counts Long/Short as separate baselines within each cell, giving 2,772 maximum distinct time-series.

13.2 Wilson Confidence Interval

The Wilson score interval is used for binomial proportion confidence intervals because it performs well with small samples and avoids boundary issues:

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

(2)

Where $\hat{p}$ is the observed hit rate, n is the sample size, and $z = 1.96$ for 95% confidence.

The **CI half-width** at key sample sizes (at $p = 0.5$):

Table 35: Wilson CI Half-Width at $p = 0.5$

n	CI Half-Width
30	17.8%
40	15.4%
50	13.9%
80	11.0%
100	9.8%
150	8.0%

13.3 Quality Gates (Corrected)

v1.1's quality gates were mathematically unattainable (e.g., requiring CI half-width $\leq 10\%$ at $n=50$ when Wilson gives 13.9%). v1.2 gates are calibrated to the Wilson table above:

Table 36: Corrected Quality Gates and Display Labels

Tier	Required n	Required CI Half-Width	Display Label
Tier 1	$n \geq 30$	$\leq 15\%$	Live Edge (Granular)
Tier 2	$n \geq 40$	$\leq 14\%$	Live Edge (Time-Conditional)
Tier 3	$n \geq 50$	$\leq 14\%$	Live Edge (Setup-Regime)
Tier 4	$n \geq 50$	$\leq 14\%$	Live Edge (Setup-Time)
Tier 5	$n \geq 80$	$\leq 12\%$	Live Edge (Setup Baseline)
Tier 6	$n \geq 50$	$\leq 14\%$	Live Edge (Global)

13.4 Multiple Comparison Correction

With hundreds of cells being tested simultaneously, false positives are inevitable. The Benjamini-Hochberg procedure controls the False Discovery Rate (FDR):

Benjamini-Hochberg FDR Control

Step 1: For all cells at a given tier, compute p-values testing H_0 : hit rate = 50% (no edge).

Step 2: Order p-values: $p(1) \leq p(2) \leq \dots \leq p(m)$

Step 3: Find largest k such that $p(k) \leq (k/m) \times \alpha$, where $\alpha = 0.10$

Step 4: Cells 1 through k are flagged as having statistically significant edge.

Step 5: Cells failing BH correction display with a "tentative" indicator and larger CI.

13.5 Bayesian Bootstrap Fallback

When no tier passes quality gates (common in early weeks), the system falls back to a Bayesian bootstrap using literature-informed priors:

Bayesian Bootstrap Prior

Prior: Beta(alpha=12, beta=8) centered at 60% hit rate with moderate confidence.

Posterior after n observations with k successes: Beta(alpha + k, beta + n - k)

Displayed as: "Bootstrap Edge (literature prior + n=[observed] observations)"

With caveat: "Insufficient direct evidence. Estimate combines prior belief with limited data."

13.6 Tier Promotion Tracking

The system tracks tier promotion events:

- When a Tier 2 cell first passes quality gates, the system logs "Tier 2 promotion: [Setup] x [Regime] x [Time] now has n=[count]"
- Dashboard displays a "Edge Maturation" panel showing which tiers are active for the current session's setups
- Weekly email summary reports tier promotion events and cells approaching quality gates

14. Runtime Layer Connection Map

Table 37: Runtime Event Triggers

Event	Frequency	Layers Triggered
Tick arrival	~1 second	Layer 9 (monitor active theses)
1-min bar close	Every minute	Layer 3 -> 4 -> 5 -> 6 -> 7 -> 8 (compact)
5-min bar close	Every 5 minutes	Layer 1 (regime recheck)
15-min bar close	Every 15 minutes	Layer 3 OI refresh
Pre-market (6:00-9:15)	Daily	Layer 1 global cues collection
Daily 8:00 AM	Daily	Layer 2 reference data refresh
Market open 9:15 AM	Daily	Layer 3 pivot/CPR computation
User expands row	On demand	Layer 8 full thesis (lazy)
14:00-15:15 IST	Daily	Layer 9 conditional expiry
15:15 IST	Daily	Layer 9 force-expiry all remaining
Nightly batch	Daily ~11 PM	Layer 10 hierarchical edge rebuild

15. Trader Daily Workflow

Pre-Market (6:00 - 9:15 AM)

Dashboard shows: expected regime from pre-market cues, F&O ban list, earnings stocks, global cue bias. No Top 25 yet (no live data).

Opening Shock (9:15 - 9:30 AM)

Regime marked "Provisional." All theses carry "Opening Shock" warning. Pre-market bias active until 9:30. Trader advised to observe, not enter.

Trend Establishment (9:30 - 10:30 AM) — PRIMARY DECISION WINDOW

Regime settles. First robust Top 25 Long + Short. Trader scans for ATTRACTIVE Net R:R, Confluence $\geq 4/6$, Tradeable tier, in leading sectors. Selects 3-5 candidates.

Mid-Morning (10:30 - 11:30 AM)

Top 25 evolves with hysteresis. New entries tagged. Trader monitors picks via live invalidation conditions. Time-decay begins for untriggered setups.

Lunch Lull (11:30 AM - 1:30 PM)

Dashboard displays "Lunch Lull — lower confidence" warning. Confluence scores typically lower. New setups possible but marked with reduced EV.

Afternoon Recovery (1:30 - 2:30 PM)

Variance recovers. New high-confidence setups can emerge. Time-decay still applies but conditional extensions activate for VWAP/Supertrend.

Square-Off (2:30 - 3:15 PM)

"Square-Off" badge displayed. New entries rare and marked with very low time-adjusted EV. At 3:00 PM, EOD force-expire warning appears.

End of Day (3:15 PM+)

All theses force-expired. Outcome Ledger complete. Trader reviews hit/miss rates. Nightly: Layer 10 rebuilds edge matrix.

16. Research Integrity and Compliance

Research-Only Product Disclaimer

This engine is a **research and information tool**, not a recommendation engine, signal service, or investment advisor. It surfaces evidence from observable market data; the trader exercises independent judgment. The engine does not predict prices, guarantee outcomes, or account for individual risk tolerance, financial situation, or investment objectives.

Validation requirements for continued honesty:

Table 38: Ongoing Validation Schedule

Component	Validation	Frequency
Confluence Score components	Correlate each check with hit rate; replace non-predictive checks	Monthly
Regime classification	Test if 3-state adds value vs global baseline	After 6 months data
Cost model	Update to latest SEBI/NSE circulars	Quarterly
Hysteresis threshold	Tune for ~1 rank change/minute target	After 30 days
Factor weights	Walk-forward optimization on out-of-sample data	Quarterly
Tier promotion	Review sufficiency as data accumulates	Monthly

17. Appendices

17.1 Appendix A: SEBI Cost Schedule Reference

All rates as of May 2026. Verify against latest NSE/SEBI circulars before deployment.

17.2 Appendix B: Setup Type Parameter Tables

Complete parameter tables for all 6 setup types are provided in Section 11.1 (Tables 21-26).

17.3 Appendix C: Glossary

Table 39: Glossary of Terms

Term	Definition
ATR	Average True Range — volatility measure
BC	Bottom Central (Central Pivot Range)
CPR	Central Pivot Range (BC, CP, TC)
EMA	Exponential Moving Average
F&O	Futures and Options
FH	First Hour (9:15-10:15 AM)
MIS	Margin Intraday Square-off (broker product)
MWPL	Market-Wide Position Limits
OI	Open Interest
ORB	Opening Range Breakout
PCR	Put-Call Ratio
PDH/PDL/PDC	Previous Day High/Low/Close
R:R	Risk-to-Reward ratio
RSI	Relative Strength Index
RV/IV	Realized Volatility / Implied Volatility ratio
SEBI	Securities and Exchange Board of India
STT	Securities Transaction Tax
TC	Top Central (Central Pivot Range)
VIX	Volatility Index (India VIX)
VWAP	Volume-Weighted Average Price

17.4 Appendix D: Change Log v1.0 to v1.2

Table 40: Changes from v1.0 to v1.2

Change	Version	Reason
8 regimes -> 3 regimes	v1.1	Overengineering without evidence
Remove FII intraday inference	v1.1	No real-time data source
Global cues pre-market only	v1.1	No signal post-9:30
Pattern recognition -> 6 mechanical checks	v1.1	Curve-fit risk
Net R:R as headline	v1.1	Trader's actual decision metric
Actionability Tier classification	v1.1	Inform-vs-restrict tension
Indian cost model explicit	v1.1	Vague costs undermined R:R
Time-decay function	v1.1	Intraday targets decay
6-tier hierarchical lookup	v1.1	Original matrix never reached sufficiency
Factor weights corrected to sum 100%	v1.2	Mathematical correction
Factor composition formulas defined	v1.2	Implementation requirement
Quality gates made mathematically attainable	v1.2	Wilson CI math correction
Mean Reversion T1 corrected for R:R ≥ 1.2	v1.2	Contradiction with Check 6
All 6 setup types formally defined	v1.2	Consistency with Layer 10
Volume seasonality adjustment	v1.2	Opening hour volume bias
Benjamini-Hochberg FDR control	v1.2	Multiple comparison correction
Setup-type-dependent expiry times	v1.2	Blanket 14:00 too aggressive
Continuous time-decay (exponential)	v1.2	Step function discontinuities
Bank Nifty divergence flag	v1.2	Dual-index regime awareness

Intraday Trading Engine — Algorithm Specification v1.2

End of Document

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